b-Divisors,
Invariant Iitaka dimension
and Nakayama Numerical Dimension

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Chapter 1

Divisors on birational models and generalised pairs

We first record divisors on every model, then introduce the nef data of a generalised pair and compare its discrepancies with ordinary discrepancies.

1.1 A divisor on all birational models

1.1A Basic notation and the category of models

All varieties are normal and defined over $\mathbb{C}$. Divisors have real coefficients unless another coefficient group is explicitly specified.

Fix a normal variety X. A *model of* X is a proper birational morphism $f : Y \to X$, with Y normal. A morphism between two such models is a proper birational morphism over X:

$$\begin{array}{ccc} Y' & \longrightarrow & Y \\ & \searrow & \downarrow f \\ & & X. \end{array}$$

Thus the category being used is the category of normal birational models over the fixed variety X.

Definition 1.1.1 (A b-divisor)**.** An *integral b-divisor* on X is an element of

$$\mathbf{Div}(X) = \varprojlim_{Y \to X} \mathrm{Div}(Y),$$

where $\mathrm{Div}(Y)$ is the group of integral Weil divisors and the transition maps are proper pushforwards. More explicitly, it is a collection

$$\mathbf{D} = (D_Y)_{Y \to X} \in \prod_{Y \to X} \mathrm{Div}(Y) \quad \text{such that} \quad g_* D_Z = D_Y \quad \text{for every } Z \xrightarrow{g} Y \to X.$$

The divisor D_Y is the *trace* of $\mathbf{D}$ on Y, also written $\mathbf{D}_Y$ or $\mathrm{tr}_Y \mathbf{D}$. Replacing $\mathrm{Div}(Y)$ by the group of rational, respectively real, Weil divisors defines rational, respectively real, b-divisors.

The compatibility can be pictured on a sequence of higher models:

$$\begin{array}{ccccccc} Y''' & \longrightarrow & Y'' & \longrightarrow & Y' & \longrightarrow & Y \\ & \searrow & & \searrow & \downarrow & \swarrow & \\ & & & & X. & & \end{array}$$

Only the commutativity over X and the pushforward compatibility are required; a b-divisor records data on every model, rather than only on one chosen sequence.

Remark 1.1.2 (Formal sums and finite traces). One sometimes writes

$$\mathbf{D} = \sum_E d_E E,$$

where E runs through the prime divisors over X, or equivalently the associated normalized divisorial valuations with centre on X. This formal sum need not be finite. On every fixed model Y, however,

$$\mathrm{tr}_Y \mathbf{D} = \sum_{E \text{ appearing as a divisor on } Y} d_E E$$

has finite support, because it is a Weil divisor on Y.

Example 1.1.3 (Information lost by pushforward)**.** Let $X = \mathbb{P}^2$ and let $f : Y = \mathrm{Bl}_p \mathbb{P}^2 \to X$ be the blowup of a point, with exceptional curve E. Let L be a line through p and let L' be its strict transform. Then

$$f_* L' = L, \qquad f_*(L' + aE) = L \quad \text{for every integer } a > 0.$$

There are therefore distinct b-divisors with traces

	X	Y
$\mathbf{D}_a$	L	$L' + aE$
$\mathbf{D}$	L	L'

For example, one may use the Cartier closures of $L' + aE$ and L', as defined below. Their traces on X coincide, but their traces on Y do not. A b-divisor retains this information on higher models.

Definition 1.1.4 (The b-divisor of a rational function)**.** For $f \in \mathbb{C}(X)^*$, set

$$\mathbf{div}_X(f) = \sum_E \mathrm{ord}_E(f) E,$$

where E runs through the prime divisors over X with centre on X. Its trace on a model Y is the usual principal divisor $\mathrm{div}_Y(f)$. Two b-divisors $\mathbf{D}_1$ and $\mathbf{D}_2$ are *linearly equivalent* if

$$\mathbf{D}_1 - \mathbf{D}_2 = \mathbf{div}_X(f)$$

for some $f \in \mathbb{C}(X)^*$.

Example 1.1.5 (The discrepancy b-divisor)**.** Consider a pair (X, B), where $B = \sum b_i B_i$ is an effective $\mathbb{Q}$-divisor and $K_X + B$ is $\mathbb{Q}$-Cartier. Choose compatible canonical divisors. The *discrepancy b-divisor* $\mathbf{A}(X, B)$ has trace $\mathbf{A}_Y$ determined by

$$K_Y = f^*(K_X + B) + \mathbf{A}_Y$$

on every model $f : Y \to X$. Here $\mathbf{A}_Y$ records the ordinary discrepancy coefficients; the log-discrepancy convention will be stated separately below.

1.1B Descent, Cartier closures, and nef data

Definition 1.1.6 (Descent)**.** A real b-divisor $\mathbf{D}$ *descends to a model* Y if $\mathbf{D}_Y$ is $\mathbb{R}$-Cartier and, on every higher model $g : Z \to Y$,

$$\mathbf{D}_Z = g^*\mathbf{D}_Y.$$

Definition 1.1.7 (Cartier closures)**.** If D is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X, its *Cartier closure* $\overline{D}$ is the b-divisor with traces

$$(\overline{D})_Y = f^*D \quad \text{on every model } f : Y \to X.$$

The same construction applies to an $\mathbb{R}$-Cartier divisor. A $\mathbb{Q}$-Cartier, respectively $\mathbb{R}$-Cartier, b-divisor is the Cartier closure of such a divisor on some model over X. If its determining divisor lies on $Y \to X$, traces on other models are obtained by pullback to a common higher model and pushforward. In particular,

$$\mathbf{div}_X(\varphi) = \overline{\operatorname{div}_X(\varphi)} \qquad (\varphi \in \mathbb{C}(X)^*).$$

Example 1.1.8. An $\mathbb{R}$-Cartier b-divisor is precisely a real b-divisor which descends to some model. Descent to X itself is an additional condition.

Definition 1.1.9 (b-nefness and NQC)**.** A b-divisor $\mathbf{D}$ is *b-nef* if it descends to a model Y on which $\mathbf{D}_Y$ is nef. It is *NQC*, meaning a nef $\mathbb{Q}$-Cartier combination, if

$$\mathbf{D} = \sum_{j=1}^{s} r_j \mathbf{D}_j, \qquad r_j \in \mathbb{R}_{\geq 0},$$

where every $\mathbf{D}_j$ is b-nef and $\mathbb{Q}$-Cartier. After clearing denominators, one can equivalently use nef Cartier b-divisors in this finite sum.

1.2 Generalized pairs with every map displayed

1.2A The data and their traces

Consider a pair (X, B) and the following data: a projective birational morphism $w : X' \to X$ and a nef $\mathbb{R}$-Cartier divisor $M_{X'}$. Put

$$M_X = w_* M_{X'},$$

and suppose that $K_X + B + M_X$ is $\mathbb{R}$-Cartier. The Cartier closure

$$\mathbf{M} = \overline{M_{X'}}$$

is the nef b-divisor whose trace on every higher model $g : Z \to X'$ is

$$M_Z = g^* M_{X'}.$$

Definition 1.2.1 (A generalized pair)**.** A *generalized pair*, or *g-pair*, is a triple $(X, B, \mathbf{M})$ consisting of the above data. In particular, B is effective, the nef b-divisor $\mathbf{M}$ descends to the specified model X', and $K_X + B + \mathbf{M}_X$ is $\mathbb{R}$-Cartier.

Remark 1.2.2 (Nefness belongs to a specified model). The definition does not require the trace to be nef on every model of X. It specifies a model X' with nef trace $M_{X'}$; all its pullbacks to models over X' are then nef. Nefness is stable under pullback, while the trace on a lower model must be considered separately. Consider the blowup

$$\pi : \mathrm{Bl}_{p,q}\,\mathbb{P}^3 \longrightarrow \mathbb{P}^3 \quad \text{with divisor} \quad H - E_1 - E_2,$$

where $p \neq q$ and H is the pullback of the hyperplane class. Let L'_{pq} be the strict transform of the line through p, q. Then

$$(H - E_1 - E_2) \cdot L'_{pq} = 1 - 1 - 1 = -1, \qquad \pi_*(H - E_1 - E_2) = [\text{hyperplane}].$$

Thus the specified divisor upstairs is not nef although its trace downstairs is nef.

Take a common model, with all maps displayed:

$$W \xrightarrow{\ g\ } X' \xrightarrow{\ w\ } X, \qquad h = w \circ g.$$

Since $\mathbf{M}$ descends to X',

$$\mathrm{tr}_W\, \mathbf{M} = M_W = g^* M_{X'}.$$

Choose compatible canonical divisors, so that, for example, $h_* K_W = K_X$, and define B_W by the equality

$$K_W + B_W + M_W = h^*(K_X + B + M_X). \tag{1.2.1}$$

Pushing forward and using $h_* M_W = M_X$ gives $h_* B_W = B$. The divisor B_W is the crepant boundary on W; it need not be effective.

1.2B Log discrepancies and the nef-data correction

Definition 1.2.3 (Log discrepancy)**.** For a prime divisor E over X, choose a sufficiently high model W on which E appears and put

$$A(E, X, B, \mathbf{M}) = 1 - \mathrm{mult}_E\, B_W.$$

The generalized pair is *log canonical*, or *lc*, if these numbers are nonnegative for every E, and *Kawamata log terminal*, or *klt*, if they are positive.

Remark 1.2.4 (Discrepancy conventions). Classical discrepancy notation has lc threshold $a = -1$. For the corresponding generalized discrepancy convention,

$$A(E, X, B, \mathbf{M}) = 1 + a(E, X, B, \mathbf{M}).$$

For an ordinary pair this reads $A(E, X, B) = 1 + a(E, X, B)$.

Remark 1.2.5 (Comparison with the ordinary pair). Suppose additionally that M_X is $\mathbb{R}$-Cartier. On a common data model $h : W \to X$, the divisor

$$F = h^* M_X - M_W$$

is exceptional over X, since $h_* F = 0$. Its negative is h-nef: the pullback $h^* M_X$ has degree zero on contracted curves, whereas M_W is nef. The negativity lemma therefore gives $F \geq 0$.[1]

[1] See [BGRrev, Lemma 5.2.17]; its discussion of real discrepancies explicitly extends the negativity lemma to real coefficients. Apply it here to the exceptional divisor F, whose negative is relatively nef.

Let B^c_W be the ordinary crepant boundary, defined by

$$K_W + B^c_W = h^*(K_X + B).$$

Comparing this equality with the generalized one gives

$$B_W = B^c_W + F.$$

Consequently generalized log discrepancies can be smaller than the ordinary log discrepancies:

$$A(E, X, B, \mathbf{M}) = A(E, X, B) - \operatorname{mult}_E F.$$

Equality holds for every E when $\mathbf{M}$ descends to X, so that $M_W = h^* M_X$.

Example 1.2.6 (A nef part which does not descend to the base model)**.** Let $X = \mathbb{P}^2$ and let $w : W = \mathrm{Bl}_p \mathbb{P}^2 \to X$ have exceptional curve E. Choose a line H not passing through p and write $H' = w^*H$. Fix $0 \le \alpha \le 1$ and set

$$B = 0, \qquad M_W = H' - \alpha E, \qquad \mathbf{M} = \overline{M_W}.$$

The b-divisor $\mathbf{M}$ descends to W. Moreover,

$$M_W = (1-\alpha)H' + \alpha(H' - E), \qquad M_X = w_* M_W = H.$$

Both H' and $H' - E$ are nef Cartier divisors; hence M_W is nef and $\mathbf{M}$ is NQC. If $\alpha > 0$, then $M_W \neq w^* M_X$, so $\mathbf{M}$ does not descend to X.

Choose compatible canonical divisors with $K_W = w^*K_X + E$. The generalized crepant equality becomes

$$w^*K_X + E + B_W + H' - \alpha E = w^*(K_X + H).$$

Therefore

$$B_W = (\alpha - 1)E, \qquad A(E, X, 0, \mathbf{M}) = 2 - \alpha.$$

The resolved sub-pair $(W, (\alpha-1)E)$ is klt, so $(X, 0, \mathbf{M})$ is klt. In particular, a negative coefficient on the crepant model is compatible with an effective boundary on the original model.

1.3 Pullbacks and restrictions

We first recall the basic definitions, following the supplied birational-geometry notes. In this section varieties are projective and normal over an algebraically closed field of characteristic zero.

Definition 1.3.1 (Real Cartier divisors and restriction)**.** A real Cartier divisor is a finite sum

$$D = \sum_i a_i D_i, \qquad a_i \in \mathbb{R}, \quad D_i \text{ Cartier}.$$

For a morphism $j : T \to Y$, pullback of its class means

$$j^*[D] = \sum_i a_i [j^* \mathcal{O}_Y(D_i)] \in \operatorname{Pic}(T) \otimes_{\mathbb{Z}} \mathbb{R}.$$

If j is an inclusion, this class is also written $D|_T$. For a complete integral curve $C \subset Y$, with normalization $\nu : C^\nu \to C$, set

$$D \cdot C = \sum_i a_i \deg_{C^\nu} \big(\nu^*(\mathcal{O}_Y(D_i)|_C)\big).$$

The divisor D is *nef* if $D \cdot C \geq 0$ for every such curve.

Remark 1.3.2 (A restricted class and a restricted divisor). For a Cartier divisor D, the line bundle $\mathcal{O}_Y(D)|_T$ always exists. If $T \subset \operatorname{Supp} D$, however, restricting local equations need not give a Cartier divisor on T. Thus restriction of the line-bundle class should be distinguished from restriction of the specified divisor. For a dominant morphism between varieties, the actual pullback of an $\mathbb{R}$-Cartier divisor is well-defined; this includes a surjective morphism.

Proposition 1.3.3 (Nefness under restriction and surjective pullback)**.** *Let D be an $\mathbb{R}$-Cartier divisor on Y.*

1. *If D is nef and $j : T \hookrightarrow Y$ is a proper subvariety, then j^*D is nef.*
2. *If $\varphi : Y' \to Y$ is proper and surjective, then D is nef if and only if φ^*D is nef.*

Proof. A curve in T is also a curve in Y, with the same degree against the restricted class. This proves (1). For (2), the projection formula gives $(\varphi^*D) \cdot C' = D \cdot \varphi_* C'$, proving the forward implication. Conversely, for every integral curve $C \subset Y$, one can cut a component of $\varphi^{-1}(C)$ dominating C by general hyperplanes to obtain a curve C' with $\varphi_* C' = dC$ for some $d > 0$. If φ^*D is nef, then $d(D \cdot C) = (\varphi^*D) \cdot C' \geq 0$. □

Chapter 2

Dimensions, abundance and surjective pullbacks

We distinguish the dimension of the images of linear systems from numerical dimension, and prove their behaviour under surjective pullback. The absolute intersection-theoretic calculations are made on projective varieties. For a proper nonprojective variety one first uses a projective birational model, by Chow's lemma and resolution, and the birational invariance proved below.

2.1 Iitaka dimension

Definition 2.1.1 (Iitaka dimension)**.** Let D be a Cartier divisor on a normal proper variety Y. Every nonempty complete linear system $|mD|$, with $m > 0$, defines a rational map

$$\varphi_{|mD|} \colon Y \dashrightarrow \mathbb{P}\big(H^0(Y, \mathcal{O}_Y(mD))^\vee\big).$$

A basis of sections gives its projective coordinates. We set

$$\kappa(Y, D) = \max_{\substack{m>0 \\ |mD|\neq\varnothing}} \dim \overline{\varphi_{|mD|}(Y)},$$

with value $-\infty$ if $|mD|$ is empty for every $m > 0$. For a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, clear denominators and use a positive Cartier multiple. The value is independent of that multiple.

Example 2.1.2. If D is an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, then $\kappa(Y, D) = \dim Y$. If $D = 0$, then

$$\kappa(Y, 0) = h^0(Y, \mathcal{O}_Y) - 1 = 0.$$

For real divisors one can use the same construction with $|\lfloor mD \rfloor|$. Its value need not be invariant under real linear equivalence.

Example 2.1.3. Let $X = \mathbb{P}^1$ and choose a rational function t with $\operatorname{div}_X(t) = [0] - [\infty]$. Put

$$D = \sqrt{2}\big([0] - [\infty]\big) = \sqrt{2}\operatorname{div}_X(t) \sim_{\mathbb{R}} 0.$$

For every positive integer m,

$$\lfloor mD \rfloor = \lfloor m\sqrt{2} \rfloor [0] - \lceil m\sqrt{2} \rceil [\infty], \qquad \deg \lfloor mD \rfloor = -1.$$

Thus $|\lfloor mD \rfloor| = \varnothing$ for every $m > 0$, and

$$\kappa(X, D) = -\infty \neq \kappa(X, 0).$$

Recall also that an integral Weil divisor F on a normal proper variety Y defines a coherent rank-one reflexive sheaf $\mathcal{O}_Y(F)$. Its sections are rational functions:

$$H^0(Y, \mathcal{O}_Y(F)) = \big\{\varphi \in \mathbb{C}(Y)^* \mid F + \operatorname{div}_Y(\varphi) \geq 0\big\} \cup \{0\}.$$

Coherence and properness imply that this is a finite-dimensional $\mathbb{C}$-vector space. The complete linear system is

$$|F| = \bigl\{F' \geq 0 \mid F' = F + \operatorname{div}_Y(\varphi) \text{ for some } \varphi \in \mathbb{C}(Y)^*\bigr\}.$$

It is nonempty if and only if $H^0(Y, \mathcal{O}_Y(F)) \neq 0$. A nonzero section φ determines the member $F + \operatorname{div}_Y(\varphi)$.

Suppose this section space is nonzero and choose a basis $\varphi_0, \dots, \varphi_r$. On a common nonempty open subset U, all these rational functions are regular and φ_0 is nowhere zero. The formula

$$y \longmapsto [\varphi_0(y) : \cdots : \varphi_r(y)]$$

defines a morphism from U to $\mathbb{P}^r$. More explicitly, it is the composition

$$\begin{aligned} &U \longrightarrow \mathbb{A}^r \hookrightarrow \mathbb{P}^r, \\ &y \longmapsto \left(\frac{\varphi_1(y)}{\varphi_0(y)}, \dots, \frac{\varphi_r(y)}{\varphi_0(y)}\right) \longmapsto [\varphi_0(y) : \cdots : \varphi_r(y)]. \end{aligned}$$

This defines the rational map $\varphi_{|F|} \colon Y \dashrightarrow \mathbb{P}^r$. Changing the basis changes it by a projective linear isomorphism of the target. The Iitaka dimension of an integral Weil divisor is defined using these maps in the same way. In particular, the floor construction makes sense even if $\lfloor mD \rfloor$ is not Cartier.

2.2 Invariant Iitaka dimension

Definition 2.2.1 (Invariant Iitaka dimension)**.** For real divisors, $D \sim_{\mathbb{R}} E$ means that

$$D - E = \sum_{j=1}^{r} a_j \operatorname{div}_Y(\varphi_j), \qquad a_j \in \mathbb{R}, \quad \varphi_j \in \mathbb{C}(Y)^*.$$

For an $\mathbb{R}$-Cartier divisor D, set

$$\kappa_\iota(Y, D) = \begin{cases} \kappa(Y, E), & D \sim_{\mathbb{R}} E \text{ for some } E \geq 0, \\ -\infty, & \text{if no such } E \text{ exists.} \end{cases}$$

If $D \geq 0$, one may take $E = D$. The following proposition proves that the value is independent of the chosen effective representative.

Proposition 2.2.2. *Let Y be a normal proper variety and let $D, E \geq 0$ be real divisors. If $D \sim_{\mathbb{R}} E$, then*

$$\kappa(Y, D) = \kappa(Y, E).$$

Proof. We first show that the Iitaka dimension of an effective real divisor depends only on its support. Suppose $A, B \geq 0$ and $\operatorname{Supp} A = \operatorname{Supp} B$. Choose positive integers a, b with

$$A \leq aB, \qquad B \leq bA.$$

The corresponding inclusions of section spaces give

$$\kappa(Y, A) \leq \kappa(Y, aB) = \kappa(Y, B), \qquad \kappa(Y, B) \leq \kappa(Y, bA) = \kappa(Y, A).$$

Here is the comparison giving the equality under multiplication. Write

$$d_m = \dim \overline{\varphi_{|\lfloor mB \rfloor|}(Y)}.$$

Then

$$\kappa(Y, B) = \max_{m \geq 1} d_m, \qquad \kappa(Y, aB) = \max_{m \geq 1} d_{am}.$$

The second maximum ranges over fewer indices, so it is at most the first. On the other hand, effectiveness gives $\lfloor mB \rfloor \leq \lfloor amB \rfloor$, hence $d_m \leq d_{am}$. This proves the reverse inequality. The same argument applies to bA. Consequently divisors with the same support have the same Iitaka dimension.

Next write

$$D - E = \sum_{j=1}^{r} a_j \operatorname{div}_Y(f_j).$$

Take the finite set of prime divisors occurring in D, E, or the $\operatorname{div}_Y(f_j)$. Equality of their coefficients is a finite linear system with integer coefficients, whose variables are the coefficients of D, E and the a_j. Impose in addition that a coefficient of D or E which is zero remains zero. The solution space is defined over $\mathbb{Q}$. Rational points are dense in its real points, and the strict positivity of each of the remaining coefficients persists in a sufficiently small neighbourhood. We can therefore choose effective rational divisors D_0, E_0 such that

$$\operatorname{Supp} D_0 = \operatorname{Supp} D, \qquad \operatorname{Supp} E_0 = \operatorname{Supp} E,$$

and

$$D_0 - E_0 = \sum_{j=1}^{r} q_j \operatorname{div}_Y(f_j), \qquad q_j \in \mathbb{Q}.$$

Choose a positive integer N clearing all the denominators. Then ND_0 and NE_0 are integral and

$$ND_0 - NE_0 = \operatorname{div}_Y\left(\prod_{j=1}^{r} f_j^{Nq_j}\right), \qquad ND_0 \sim NE_0.$$

Combining ordinary invariance under linear equivalence with the support comparison just proved gives

$$\begin{aligned}\kappa(Y, D) &= \kappa(Y, D_0) = \kappa(Y, ND_0) \\ &= \kappa(Y, NE_0) = \kappa(Y, E_0) = \kappa(Y, E).\end{aligned}$$

Thus the definition of κ_ι is independent of E. □

2.3 Numerical dimension of a nef divisor

Definition 2.3.1 (Numerical dimension)**.** Let D be a nef $\mathbb{R}$-Cartier divisor on Y, let $n = \dim Y$, and let A be an ample Cartier divisor. Set

$$\nu(Y, D) = \max\bigl\{0 \leq k \leq n \mid D^k \cdot A^{n-k} > 0\bigr\}.$$

This is the numerical dimension of D.

Proposition 2.3.2. *In the preceding setting, the following assertions hold.*

1. *Equivalently,*

$$\nu(Y,D) = \max\bigl\{\dim T \mid T \subseteq Y \text{ an integral subvariety, } D^{\dim T}\cdot T > 0\bigr\}.$$

2. *The value of $\nu(Y,D)$ is independent of the choice of the ample Cartier divisor A.*

Proof. For (1), first suppose $D^k \cdot A^{n-k} > 0$. Choose $m \gg 0$ such that mA is very ample. General members $H_1, \dots, H_{n-k} \in |mA|$ meet properly and give an effective k-cycle

$$Z = H_1 \cdots H_{n-k} \cdot [Y] = \sum_i c_i [T_i], \qquad c_i > 0, \quad \dim T_i = k.$$

Consequently

$$0 < m^{n-k}\bigl(D^k \cdot A^{n-k}\bigr) = D^k \cdot Z = \sum_i c_i \bigl(D^k \cdot T_i\bigr).$$

Every summand is nonnegative because D is nef. Hence at least one T_i satisfies $D^k \cdot T_i > 0$.

Conversely, let $T \subseteq Y$ be an integral subvariety of dimension k with $D^k \cdot T > 0$. For $m \gg 0$, the sheaf $\mathcal{I}_T \otimes \mathcal{O}_Y(mA)$ is globally generated. Thus the linear system of sections of mA vanishing along T has base locus T. Choose $n-k$ general members $H_1, \dots, H_{n-k}$ of this system. They meet properly and their intersection has the form

$$Z = H_1 \cdots H_{n-k} \cdot [Y] = c[T] + R, \qquad c > 0,$$

where R is an effective k-cycle. It follows that

$$\begin{aligned} m^{n-k}\bigl(D^k \cdot A^{n-k}\bigr) &= D^k \cdot Z \\ &= c\bigl(D^k \cdot T\bigr) + D^k \cdot R \\ &\geq c\bigl(D^k \cdot T\bigr) > 0. \end{aligned}$$

Therefore $D^k \cdot A^{n-k} > 0$. If $k = n$, then $T = Y$ and the assertion is immediate.

For (2), let A, H be ample Cartier divisors. Choose one positive integer m sufficiently large that both $mA - H$ and $mH - A$ are ample. For $k < n$, put $P = mA - H$. Expanding $(H+P)^{n-k}$ and using nonnegativity of intersections of nef divisors gives

$$m^{n-k}\bigl(D^k \cdot A^{n-k}\bigr) = D^k \cdot (H+P)^{n-k} \geq D^k \cdot H^{n-k}.$$

Interchanging A and H gives

$$m^{n-k}\bigl(D^k \cdot H^{n-k}\bigr) \geq D^k \cdot A^{n-k}.$$

Thus

$$D^k \cdot A^{n-k} > 0 \quad \Longleftrightarrow \quad D^k \cdot H^{n-k} > 0.$$

For $k = n$ the expression is D^n and does not involve either ample divisor. This proves independence. □

Example 2.3.3. Let X be a smooth projective surface and let D be a nef $\mathbb{R}$-Cartier divisor. Then

$$\begin{aligned} &\nu(X,D) = 2 \quad \text{if} \quad D^2 > 0, \\ &\nu(X,D) = 1 \quad \text{if} \quad D^2 = 0 \text{ and } D \not\equiv 0, \\ &\nu(X,D) = 0 \quad \text{if} \quad D \equiv 0. \end{aligned}$$

2.4 Sigma dimension

Definition 2.4.1 (Sigma dimension)**.** Let Y be smooth and projective and let D be an $\mathbb{R}$-Cartier divisor. For a Cartier divisor A, put

$$\sigma(D;A)=\max\left\{k\in\mathbb{Z}_{\geq 0}\;\middle|\;\limsup_{m\to\infty}\frac{h^0\big(Y,\mathcal{O}_Y(\lfloor mD\rfloor+A)\big)}{m^k}>0\right\}.$$

An empty set has value $-\infty$. Define

$$\kappa_\sigma(Y,D)=\max_A\sigma(D;A),$$

where A ranges over Cartier divisors. If Y is normal, take a smooth projective resolution and use the pullback of D.

Proposition 2.4.2. *The definition of $\kappa_\sigma(Y,D)$ is independent of the chosen smooth model. If D is nef, then*

$$\kappa_\sigma(Y,D)=\nu(Y,D).$$

Proof. The second assertion is Nakayama's comparison theorem for nef divisors; see [Nak04, Chapter V] and the dimension conventions recalled in [Hu20, Section 2.3]. We prove the first assertion.

Let $f\colon Z\to W$ be a birational morphism of smooth projective models, and let D be an $\mathbb{R}$-Cartier divisor on W. Set

$$E_m=\lfloor mf^*D\rfloor-f^*\lfloor mD\rfloor=\big\lfloor f^*\{mD\}\big\rfloor.$$

This is effective and f-exceptional: the fractional part is effective, and no nonexceptional component acquires an integral coefficient when pulled back. Normality of W therefore gives

$$f_*\mathcal{O}_Z(E_m)=\mathcal{O}_W.$$

For every Cartier divisor A on W, the projection formula gives

$$\begin{aligned}&H^0\big(Z,\mathcal{O}_Z(\lfloor mf^*D\rfloor+f^*A)\big)\\&\quad=H^0\big(W,f_*\mathcal{O}_Z(E_m+f^*\lfloor mD\rfloor+f^*A)\big)\\&\quad\cong H^0\big(W,\mathcal{O}_W(\lfloor mD\rfloor+A)\big).\end{aligned}$$

Comparing growth gives

$$\kappa_\sigma(W,D)\leq\kappa_\sigma(Z,f^*D),$$

since Cartier divisors on Z include the pullbacks of those on W.

For the other inequality, fix a Cartier divisor B on Z and an ample Cartier divisor A_0 on W. The sheaf $f_*\mathcal{O}_Z(-B)$ is coherent of rank one, so for $t\gg 0$ it has a nonzero section after tensoring with $\mathcal{O}_W(tA_0)$. By the projection formula,

$$\begin{aligned}&H^0\big(Z,\mathcal{O}_Z(tf^*A_0-B)\big)\\&\quad=H^0\big(W,f_*(\mathcal{O}_Z(-B)\otimes f^*\mathcal{O}_W(tA_0))\big)\\&\quad\cong H^0\big(W,f_*\mathcal{O}_Z(-B)\otimes\mathcal{O}_W(tA_0)\big)\neq 0.\end{aligned}$$

Choose such a nonzero section once, with t fixed independently of m. Multiplication by it gives an injection

$$H^0\big(Z,\mathcal{O}_Z(\lfloor mf^*D\rfloor+B)\big)\hookrightarrow H^0\big(Z,\mathcal{O}_Z(\lfloor mf^*D\rfloor+f^*(tA_0))\big).$$

The equality proved in the first part then yields

$$\begin{aligned} h^0\big(Z, \mathcal{O}_Z(\lfloor mf^*D \rfloor + B)\big) &\leq h^0\big(Z, \mathcal{O}_Z(\lfloor mf^*D \rfloor + f^*(tA_0))\big) \\ &= h^0\big(W, \mathcal{O}_W(\lfloor mD \rfloor + tA_0)\big). \end{aligned}$$

Consequently

$$\sigma_Z(f^*D; B) \leq \sigma_W(D; tA_0) \leq \kappa_\sigma(W, D).$$

Taking the maximum over B gives $\kappa_\sigma(Z, f^*D) \leq \kappa_\sigma(W, D)$. This proves equality. Two resolutions are compared by taking a common smooth resolution, so the definition is independent of the chosen model. □

2.5 Abundant divisors

Definition 2.5.1 (Abundant divisor)**.** An $\mathbb{R}$-Cartier divisor D is abundant if

$$\kappa_\iota(Y, D) = \kappa_\sigma(Y, D).$$

In particular,

$$D \text{ is nef and abundant} \quad \Longleftrightarrow \quad D \text{ is nef and } \kappa_\iota(Y, D) = \nu(Y, D).$$

For a $\mathbb{Q}$-divisor this is the usual equality $\kappa(Y, D) = \nu(Y, D)$.

Example 2.5.2 (A nef divisor which is not abundant)**.** Let C be an elliptic curve and let $L \in \mathrm{Pic}^0(C)$ be non-torsion: no positive tensor power of L is trivial. Choose an integral divisor D with $L \cong \mathcal{O}_C(D)$. Since $\deg D = 0$, the divisor D is nef and $\nu(C, D) = 0$.

If there were a nonzero section of $\mathcal{O}_C(mD)$ for some $m > 0$, its zero divisor would be effective of degree zero. It would therefore be empty, so the section would be nowhere vanishing and would give

$$\mathcal{O}_C \cong L^{\otimes m},$$

contrary to the choice of L. Hence $H^0(C, \mathcal{O}_C(mD)) = 0$ for all $m > 0$, and $\kappa(C, D) = -\infty$. The invariant Iitaka dimension is also $-\infty$.[1] Thus D is not abundant.

Example 2.5.3. An ample $\mathbb{R}$-Cartier divisor and the zero divisor are both nef and abundant.

Lemma 2.5.4. *Let $f\colon Y' \to Y$ be a proper surjective morphism and let D be a nef $\mathbb{R}$-Cartier divisor on Y. Then*

$$\nu(Y', f^*D) = \nu(Y, D).$$

Proof. Use the equivalent definition in Proposition 2.3.2(1). For an integral subvariety $W \subseteq Y'$ of dimension k, the projection formula gives

$$(f^*D)^k \cdot W = D^k \cdot f_*[W].$$

[1] To justify the last assertion directly, an effective real divisor real-linearly equivalent to D has degree zero and must be zero. An integral divisor which is real-principal is rational-principal: after fixing the finitely many rational functions in its real-principal presentation, the coefficient equations form a consistent rational linear system and have a rational solution. Clearing denominators would make a positive multiple of D principal, and thus make L torsion.

If $\dim f(W) < k$, then $f_*[W] = 0$ as a k-cycle. Otherwise

$$f_*[W] = d[f(W)], \qquad d = [\mathbb{C}(W) : \mathbb{C}(f(W))] > 0.$$

Thus positivity of $(f^*D)^k \cdot W$ implies positivity of $D^k \cdot f(W)$, and

$$\nu(Y', f^*D) \leq \nu(Y, D).$$

Conversely, let $T \subseteq Y$ be an integral subvariety of dimension k with $D^k \cdot T > 0$. Take a closed point of the generic fibre of $f^{-1}(T) \to T$, and let W be its closure in Y' with the reduced scheme structure. The residue field of that closed point is a finite extension of $\mathbb{C}(T)$. Hence

$$[\mathbb{C}(W) : \mathbb{C}(T)] < \infty,$$

so $W \to T$ is generically finite and $\dim W = k$. Applying the projection formula again gives

$$(f^*D)^k \cdot W = \deg(W \to T)\big(D^k \cdot T\big) > 0.$$

Therefore $\nu(Y, D) \leq \nu(Y', f^*D)$, proving equality. □

Lemma 2.5.5. *For a proper surjective morphism $f : Y' \to Y$ and an $\mathbb{R}$-Cartier divisor D on Y, there exists an effective divisor $E \sim_{\mathbb{R}} D$ if and only if there exists an effective divisor $E' \sim_{\mathbb{R}} f^*D$.*

Proof. The only-if part is immediate. We prove the if part. Write

$$f^*D - E' = \sum_j r_j \operatorname{div}(f_j), \qquad r_j \in \mathbb{R}, \quad f_j \in \mathbb{C}(Y')^*.$$

The effective $\mathbb{R}$-Cartier divisor E' can be written as

$$E' = \sum_\ell \lambda_\ell E_\ell, \qquad \lambda_\ell > 0,$$

where each E_ℓ is an effective Cartier divisor.[2] Let η be the generic point of Y. The generic fibre

$$Y'_\eta = Y' \times_Y \operatorname{Spec} k(\eta)$$

is an integral scheme over $k(\eta)$. Choose a closed point $x \in Y'_\eta$ outside the supports of the E_ℓ and the zero and pole divisors of all the f_j. Let $W \subseteq Y'$ be the closure of x, with its reduced scheme structure, and let W^ν be its normalisation. Write

$$\tau : \overline{T} = W^\nu \longrightarrow W \hookrightarrow Y', \qquad h = f \circ \tau : \overline{T} \longrightarrow Y.$$

Then h is proper and generically finite of degree $d = [k(\overline{T}) : k(\eta)] < \infty$.

By the choice of x, each f_j pulls back to a nonzero rational function $g_j = \tau^* f_j \in \mathbb{C}(\overline{T})^*$. Moreover, each $\tau^* E_\ell$ is effective, so $\tau^* E' \geq 0$. Therefore

$$h^*D - \tau^*E' = \tau^*(f^*D - E') = \sum_j r_j \operatorname{div}_{\overline{T}}(g_j).$$

Proper pushforward of a principal divisor is given by the field norm:

$$h_* \operatorname{div}_{\overline{T}}(g_j) = \operatorname{div}_Y \operatorname{Nm}_{\mathbb{C}(\overline{T})/\mathbb{C}(Y)}(g_j);$$

[2] See Fujino [Fuj, Lemma 0.14].

see [Stacks, Tag 02RT] or [Ful98, §1.4]. In particular, proper pushforward preserves $\mathbb{R}$-linear equivalence. Thus

$$h_* h^* D = dD \sim_{\mathbb{R}} h_* \tau^* E' \geq 0.$$

Dividing by d gives the required effective representative of D. □

We also need a simple fact.

Lemma 2.5.6. *Let $E \geq 0$ be an $\mathbb{R}$-divisor on a normal proper variety Y, and put $U = Y \setminus \operatorname{Supp} E$. Inside $\mathbb{C}(Y)$,*

$$H^0(U, \mathcal{O}_U) = \bigcup_{m \geq 0} H^0\big(Y, \mathcal{O}_Y(\lfloor mE \rfloor)\big).$$

Moreover,

$$\kappa(Y, E) = \operatorname{tr.deg}_{\mathbb{C}} \operatorname{Frac} H^0(U, \mathcal{O}_U).$$

Proof. A rational function on the right-hand side has no poles outside $\operatorname{Supp} E$. It is therefore regular in codimension one on U, and normality implies that it is regular on U. Conversely, a regular function on U has poles only along the finitely many prime components of E; their orders are bounded by $\lfloor mE \rfloor$ for sufficiently large m. This proves the first equality.

Since $E \geq 0$, the constant function 1 belongs to $H^0(Y, \mathcal{O}_Y(\lfloor mE \rfloor))$ for every m. Choose a basis

$$\{1, f_{m,1}, \ldots, f_{m,N_m}\}$$

of this vector space. The corresponding rational map restricts to a morphism

$$\begin{aligned} \overline{\Phi}_m : U &\longrightarrow \mathbb{P}^{N_m}, \\ y &\longmapsto [1 : f_{m,1}(y) : \cdots : f_{m,N_m}(y)]. \end{aligned}$$

Its image lies in an affine chart. Let Z_m be the closure of that image, and put

$$K_m = \mathbb{C}(f_{m,1}, \ldots, f_{m,N_m}) \subseteq \mathbb{C}(Y).$$

Then

$$\dim Z_m = \operatorname{tr.deg}_{\mathbb{C}} K_m, \qquad \kappa(Y, E) = \max_m \operatorname{tr.deg}_{\mathbb{C}} K_m.$$

The inclusion $\lfloor mE \rfloor \leq \lfloor (m+1)E \rfloor$ gives $K_m \subseteq K_{m+1}$. Consequently

$$\operatorname{Frac} H^0(U, \mathcal{O}_U) = \bigcup_m K_m,$$

which proves the second equality. □

Proposition 2.5.7. *For a proper surjective morphism $g : Y' \to Y$ between normal proper varieties and an $\mathbb{R}$-Cartier divisor D on Y, one has*

$$\kappa_\iota(Y', g^* D) = \kappa_\iota(Y, D).$$

Proof. If D has no effective $\mathbb{R}$-linearly equivalent representative, Lemma 2.5.5 shows that neither does $g^* D$, so both sides are $-\infty$. Otherwise choose $D \sim_{\mathbb{R}} E \geq 0$. It suffices to compare $\kappa(Y', g^* E)$ and $\kappa(Y, E)$.

The divisor E is $\mathbb{R}$-Cartier, and

$$\operatorname{Supp}(g^*E) = g^{-1}(\operatorname{Supp} E).$$

Put $U = Y \setminus \operatorname{Supp} E$ and $V = g^{-1}(U)$. Take the Stein factorisation

$$g : Y' \xrightarrow{a} \widetilde{Y} \xrightarrow{b} Y, \qquad a_*\mathcal{O}_{Y'} = \mathcal{O}_{\widetilde{Y}}, \quad b \text{ finite.}$$

Then

$$H^0(V, \mathcal{O}_V) = H^0(b^{-1}U, \mathcal{O}_{\widetilde{Y}}),$$

and the latter ring is integral over $H^0(U, \mathcal{O}_U)$.[3] Lemma 2.5.6 now gives

$$\begin{aligned} \kappa(Y', g^*E) &= \operatorname{tr.deg}_{\mathbb{C}} \operatorname{Frac} H^0(V, \mathcal{O}_V) \\ &= \operatorname{tr.deg}_{\mathbb{C}} \operatorname{Frac} H^0(b^{-1}U, \mathcal{O}_{\widetilde{Y}}) \\ &= \operatorname{tr.deg}_{\mathbb{C}} \operatorname{Frac} H^0(U, \mathcal{O}_U) \\ &= \kappa(Y, E). \end{aligned}$$

This proves the assertion. □

Proposition 2.5.8 (Surjective pullback of nef abundance)**.** *Let $g : Y' \to Y$ be a proper surjective morphism between normal projective varieties, and let D be an $\mathbb{R}$-Cartier divisor on Y. Then*

$$D \text{ is nef and abundant} \quad \Longleftrightarrow \quad g^*D \text{ is nef and abundant.}$$

Proof. Nefness is preserved by and descends under surjective pullback. For abundance, Lemma 2.5.4 and Proposition 2.5.7 give

$$\kappa_\iota(Y, D) = \kappa_\iota(Y', g^*D), \qquad \nu(Y, D) = \nu(Y', g^*D).$$

The required equivalence follows. □

2.6 Relative dimensions

Definition 2.6.1 (The relative version)**.** Let $f : X \to Z$ be a proper morphism from a normal algebraic variety, and let D be an $\mathbb{R}$-Cartier divisor on X. The *relative numerical dimension* of D over Z is

$$\kappa_\sigma(X/Z, D) = \kappa_\sigma(\overline{F}, D|_{\overline{F}}),$$

[3] This is immediate when U is affine. For general U, put $A = H^0(U, \mathcal{O}_U)$ and $B = H^0(b^{-1}U, \mathcal{O}_{\widetilde{Y}})$. We must show that every $\beta \in B$ satisfies a monic polynomial with coefficients in A. Let $K = \mathbb{C}(Y)$ and $L = \mathbb{C}(\widetilde{Y})$. Since b is finite and dominant, L/K is finite. Let $F_\beta(T) = T^n + c_{n-1}T^{n-1} + \cdots + c_0 \in K[T]$ be the monic minimal polynomial of β over K. On every affine open $W_i = \operatorname{Spec} R_i \subseteq U$, finiteness of b shows that the restriction of β to $b^{-1}W_i$ is integral over R_i; let $Q_i(T) \in R_i[T]$ be a monic polynomial annihilating it. Then $F_\beta \mid Q_i$ in $K[T]$. Indeed, polynomial division gives $Q_i = H_iF_\beta + S_i$ with $\deg S_i < \deg F_\beta$, and $S_i(\beta) = 0$ forces $S_i = 0$ by minimality. In an algebraic closure of K, write $F_\beta(T) = \prod_j (T - \beta_j)$. Every β_j is a root of Q_i and hence integral over R_i. Each c_j is, up to sign, an elementary symmetric polynomial in the β_j, so it is integral over R_i. Normality of Y makes R_i integrally closed in $K = \operatorname{Frac} R_i$, whence $c_j \in R_i$ for every i. These coefficients therefore belong to A, proving that β is integral over A.

where $\overline{F}$ is a geometric generic fibre, with the Stein factorisation understood.[4] The *relative invariant Iitaka dimension* is

$$\kappa_\iota(X/Z, D) = \kappa_\iota(\overline{F}, D|_{\overline{F}}).$$

We call D *abundant over* Z if

$$\kappa_\iota(X/Z, D) = \kappa_\sigma(X/Z, D).$$

Remark 2.6.2. By [Via13, Lemma 2.1], a very general fibre and the geometric generic fibre are isomorphic as abstract schemes: the isomorphism is taken over a field isomorphism $\mathbb{C} \simeq \overline{k(\eta)}$.

Proposition 2.6.3 (Relative surjective pullback)**.** *Let $f : X \to Z$ be a proper morphism from a normal algebraic variety to a variety, let $g : Y \to X$ be a proper surjective morphism of normal varieties, and let D be an $\mathbb{R}$-Cartier divisor on X. Then D is nef and abundant over Z if and only if g^*D is nef and abundant over Z; compare [Hu20, Lemma 2.10].*

Proof. Take the Stein factorisations

$$X \xrightarrow{a} S_X \xrightarrow{\pi} Z, \qquad Y \xrightarrow{b} S_Y \xrightarrow{\rho} Z.$$

After shrinking Z, we may assume that the finite morphisms π and ρ are étale. Their fibres are disjoint unions of spectra of finite separable field extensions. The closed subsets avoided by very general fibres of a and b have proper closed images in Z: use properness and the dimension property of the finite étale morphisms π and ρ. Choose a very general closed point $z \in Z$ outside the union of these images. Then

$$X_z = \coprod_{s \in \pi^{-1}(z)} X_s, \qquad Y_z = \coprod_{t \in \rho^{-1}(z)} Y_t.$$

By the preceding remark, each X_s and Y_t is isomorphic as an abstract scheme to the corresponding geometric generic fibre of a or b, over a generic point of S_X or S_Y. We may therefore compare the dimensions on fixed components X_s and Y_t.

Fix an integral component X_s. The proper surjection $g_z : Y_z \to X_z$ shows that the finitely many closed images $g_z(Y_t)$ cover X_s. Since X_s is irreducible, at least one of these images equals X_s. Choose such a t. The corresponding restriction is a proper surjective morphism

$$g_{t,s} : Y_t \longrightarrow X_s.$$

Proposition 2.5.8 applies to this morphism and gives the desired equality of dimensions. Relative nefness itself is preserved by and descends under surjective pullback, by the projection formula for curves contracted over Z. This proves the result. □

[4] First replace Z by the image of f. If the geometric generic fibre of $X \to Z$ is disconnected, take the Stein factorisation and use a geometric generic fibre of its contraction. Over a geometric point, the components correspond to the branches of the finite Stein factor. The proof below makes this componentwise interpretation explicit.

Chapter 3

Asymptotic vanishing orders

3.1 The asymptotic vanishing order

Definition 3.1.1 (Asymptotic vanishing order)**.** Let P be a prime divisor over a normal proper variety X, with associated discrete valuation mult_P. For an $\mathbb{R}$-Cartier divisor D on X, define

$$o_P(D) = \inf\{\mathrm{mult}_P G \mid D \sim_{\mathbb{R}} G \geq 0\}.$$

We write

$$|D|_{\mathbb{R}} = \{G \geq 0 \mid G \sim_{\mathbb{R}} D\}$$

for the effective real linear system. The infimum of the empty set is $+\infty$. If P is not on X, the multiplicity is computed after pulling back G to a model containing P.

Lemma 3.1.2. *In the above situation, suppose that the real linear systems in question are nonempty. Then:*

1. *If $D \sim_{\mathbb{R}} D'$, then $o_P(D) = o_P(D')$.*
2. *If $a > 0$, then $o_P(aD) = a\, o_P(D)$.*
3. *We have*
$$o_P(D_1 + D_2) \leq o_P(D_1) + o_P(D_2).$$
4. *If $E \geq 0$ is an $\mathbb{R}$-Cartier divisor, then*
$$o_P(D + E) \leq o_P(D) + \mathrm{mult}_P E.$$

Proof. (1) follows from $|D'|_{\mathbb{R}} = |D|_{\mathbb{R}}$.

For (2), we have $\mathrm{mult}_P(aG) = a\,\mathrm{mult}_P G$, and multiplication by a is a bijection from $|D|_{\mathbb{R}}$ to $|aD|_{\mathbb{R}}$.

For (3), given $\delta > 0$, choose $G_i \in |D_i|_{\mathbb{R}}$ such that

$$\mathrm{mult}_P G_i < o_P(D_i) + \delta \qquad (i = 1, 2).$$

Then $G_1 + G_2 \in |D_1 + D_2|_{\mathbb{R}}$, and

$$\begin{aligned} o_P(D_1 + D_2) &\leq \mathrm{mult}_P(G_1 + G_2) \\ &= \mathrm{mult}_P G_1 + \mathrm{mult}_P G_2 \\ &< o_P(D_1) + o_P(D_2) + 2\delta. \end{aligned}$$

Letting $\delta \to 0$ gives the assertion.

For (4), $D + E \sim_{\mathbb{R}} G + E$ for every $G \in |D|_{\mathbb{R}}$, and $G + E$ is effective. Hence

$$o_P(D + E) \leq \mathrm{mult}_P(G + E) = \mathrm{mult}_P G + \mathrm{mult}_P E.$$

Taking the infimum over $G \in |D|_{\mathbb{R}}$ proves (4). □

3.2 Numerical asymptotic vanishing orders

Definition 3.2.1 (Numerical asymptotic vanishing order)**.** Let X be a normal projective variety, let D be a pseudo-effective $\mathbb{R}$-Cartier divisor, and let A be an ample $\mathbb{R}$-Cartier divisor. For a prime divisor P over X, the *numerical asymptotic vanishing order* is

$$\sigma_P(D) = \lim_{\epsilon \downarrow 0} o_P(D + \epsilon A).$$

Lemma 3.2.2. *The definition is independent of the choice of A and depends only on the numerical class of D.*

Proof. For an ample $\mathbb{R}$-Cartier divisor H, one has $o_P(H) = 0$. Indeed, positive combinations of general members of basepoint-free multiples can be chosen to avoid the centre of P. By Lemma 3.1.2(3),

$$o_P(L + H) \leq o_P(L)$$

whenever L is an $\mathbb{R}$-Cartier divisor with $|L|_{\mathbb{R}} \neq \varnothing$. For $\epsilon > 0$, the divisor $D + \epsilon A$ is big, so its effective real linear system is nonempty. In particular, if $0 < \epsilon_1 < \epsilon_2$, then

$$o_P(D + \epsilon_2 A) \leq o_P(D + \epsilon_1 A).$$

Thus $o_P(D+\epsilon A)$ is nondecreasing as $\epsilon \downarrow 0$. To see that the limit is finite, take a projective model $\mu : Z \to X$ containing P, an ample Cartier divisor H on Z, and put $n = \dim X$. For every $G \in |D + \epsilon A|_{\mathbb{R}}$, the divisor $\mu^* G$ is effective, so

$$0 \leq \operatorname{mult}_P G \leq \frac{\mu^*(D + \epsilon A) \cdot H^{n-1}}{P \cdot H^{n-1}}.$$

The denominator is positive, and the right-hand side is uniformly bounded for $0 < \epsilon \leq 1$. Hence the defining limit exists and is finite.

Suppose that A and B are ample $\mathbb{R}$-Cartier divisors. Choose $c \gg 0$ such that $cB - A$ is ample. Then

$$D + c\epsilon B = (D + \epsilon A) + \epsilon(cB - A),$$

and the preceding inequality gives

$$o_P(D + c\epsilon B) \leq o_P(D + \epsilon A).$$

For the moment, denote by σ_P^A and σ_P^B the limits using A and B, respectively. Letting $\epsilon \downarrow 0$ gives

$$\sigma_P^B(D) \leq \sigma_P^A(D).$$

Interchanging A and B proves

$$\sigma_P^A(D) = \sigma_P^B(D),$$

which proves the first assertion.

Now suppose that $D \equiv D'$. For every $\epsilon > 0$, set

$$H_\epsilon = D' - D + \epsilon A \equiv \epsilon A.$$

This is an ample $\mathbb{R}$-Cartier divisor. Since

$$D' + 2\epsilon A = (D + \epsilon A) + H_\epsilon,$$

we have

$$\begin{aligned} o_P(D' + 2\epsilon A) &= o_P\big((D + \epsilon A) + H_\epsilon\big) \\ &\leq o_P(D + \epsilon A). \end{aligned}$$

Letting $\epsilon \downarrow 0$ gives $\sigma_P(D') \leq \sigma_P(D)$. Interchanging D and D' gives the reverse inequality. Therefore $\sigma_P(D') = \sigma_P(D)$. □

Proposition 3.2.3. *If D is a nef $\mathbb{R}$-Cartier divisor, then $\sigma_P(D) = 0$ for every prime divisor P over X.*

Proof. Fix an ample divisor A. For every $\epsilon > 0$, the $\mathbb{R}$-Cartier divisor $D + \epsilon A$ is ample. Take a model $\mu : Y \to X$ containing P, and let $C = \mu(P) \subseteq X$ be its centre. Write

$$D + \epsilon A = \sum_i r_i H_i, \qquad r_i > 0,$$

where the H_i are ample Cartier divisors. For each i, choose $m_i \gg 0$ such that $|m_i H_i|$ is basepoint-free. We can then choose

$$Z_i \in |m_i H_i|, \qquad C \not\subseteq \operatorname{Supp} Z_i.$$

Consequently

$$\operatorname{mult}_P Z_i = \operatorname{coeff}_P(\mu^* Z_i) = 0.$$

Set

$$G_\epsilon = \sum_i \frac{r_i}{m_i} Z_i \in |D + \epsilon A|_{\mathbb{R}}.$$

This is effective and $\operatorname{mult}_P G_\epsilon = 0$. By definition,

$$0 \leq o_P(D + \epsilon A) \leq \operatorname{mult}_P G_\epsilon = 0$$

for every $\epsilon > 0$. Hence

$$\sigma_P(D) = \lim_{\epsilon \downarrow 0} o_P(D + \epsilon A) = 0.$$

□

Example 3.2.4. Let E be a smooth elliptic curve and let $L \in \operatorname{Pic}^0(E)$ be a non-torsion line bundle, so that $L^{\otimes m} \ncong \mathcal{O}_E$ for every positive integer m. Choose a Cartier divisor D such that $L \cong \mathcal{O}_E(D)$. Then D is numerically trivial and hence nef. For any divisor A of positive degree, Riemann–Roch gives

$$\begin{aligned} &h^0\big(E, \mathcal{O}_E(mD + A)\big) - h^0\big(E, \mathcal{O}_E(K_E - mD - A)\big) \\ &\quad = h^0\big(E, \mathcal{O}_E(mD + A)\big) \\ &\quad = \deg(mD + A) = \deg A \end{aligned}$$

for every integer m. In fact, the second term in this display is zero because $\deg(K_E - mD - A) = -\deg A < 0$. If $\deg A \leq 0$, then $h^0(E, \mathcal{O}_E(mD + A)) \leq 1$. Thus for every fixed A these section spaces are bounded independently of m, whereas positive-degree A gives a positive constant. Consequently

$$\kappa_\sigma(E, D) = \max_A \sigma(D; A) = 0.$$

If there were an effective $\mathbb{R}$-Cartier divisor $G \sim_{\mathbb{R}} D$, then $\deg G = 0$ would force $G = 0$. Hence $D \sim_{\mathbb{R}} 0$, so write

$$D = \sum_i r_i \operatorname{div}(f_i), \qquad r_i \in \mathbb{R}, \quad f_i \in \mathbb{C}(E)^*.$$

Let $Q_1, \dots, Q_s$ be all the points occurring in the support of D or in the divisors $\operatorname{div}(f_i)$, and write

$$D = \sum_{k=1}^{s} d_k Q_k, \qquad \operatorname{div}(f_i) = \sum_{k=1}^{s} n_{ki} Q_k.$$

Comparing coefficients gives the finite linear system

$$\sum_i n_{ki} r_i = d_k \qquad (k = 1, \dots, s), \qquad n_{ki}, d_k \in \mathbb{Z}.$$

Rational solutions are dense in the real solution space of this consistent rational system. We can therefore choose $q_i \in \mathbb{Q}$ with

$$D = \sum_i q_i \operatorname{div}(f_i),$$

that is, $D \sim_{\mathbb{Q}} 0$. Choose a positive integer N clearing the denominators of the q_i. Then

$$ND = \operatorname{div}\left(\prod_i f_i^{Nq_i}\right), \qquad L^{\otimes N} \cong \mathcal{O}_E,$$

contrary to non-torsion. Therefore $|D|_{\mathbb{R}} = \varnothing$, and

$$\kappa_\iota(E, D) = -\infty, \qquad o_P(D) = +\infty.$$

On the other hand, $\sigma_P(D) = 0$ because D is nef.

3.3 Nakayama–Zariski decompositions

Definition 3.3.1 (Nakayama–Zariski decomposition)**.** Let D be a pseudo-effective $\mathbb{R}$-divisor on a smooth projective variety X. Set

$$N_\sigma(D) = \sum_{\Gamma \subset X} \sigma_\Gamma(D)\Gamma, \qquad P_\sigma(D) = D - N_\sigma(D),$$

where the sum runs over prime divisors Γ on X. By [Nak04, Chapter III, Corollary 1.11], there are only finitely many prime divisors Γ with $\sigma_\Gamma(D) > 0$, so this is a finite sum. The decomposition

$$D = P_\sigma(D) + N_\sigma(D)$$

is called the *σ-decomposition* of D. The divisors $P_\sigma(D)$ and $N_\sigma(D)$ are its *positive part* and *negative part*, respectively. If $P_\sigma(D)$ is nef, this is called a *Zariski decomposition*.

Theorem 3.3.2 (Zariski decomposition on a smooth projective surface)**.** *Let X be a smooth projective surface, and let D be a pseudo-effective integral divisor on X. Then D can be written uniquely as a sum*

$$D = P + N$$

of $\mathbb{Q}$-divisors with the following properties:

(i) *P is nef.*

(ii) *$N = \sum_i a_i E_i$ is effective, where the E_i are distinct prime curves and $a_i > 0$. If $N \neq 0$, the intersection matrix $(E_i \cdot E_j)$ is negative definite.*

(iii) *P is orthogonal to each component of N:*

$$P \cdot E_i = 0 \qquad \text{for every } i.$$

Zariski established this decomposition theorem for effective divisors in [Zar62]; Fujita extended it to pseudo-effective divisors in [Fuj79].

Example 3.3.3. Consider the blowup

$$\pi : X = \mathrm{Bl}_p \mathbb{P}^2 \longrightarrow \mathbb{P}^2,$$

and set $D = H + E$, where E is the exceptional curve and H is the pullback of a line. Then

$$P = H, \qquad N = E,$$

with P nef, $P \cdot E = 0$, and $E^2 = -1$. These are the positive and negative parts of the Zariski decomposition.

The divisor $A = 2H - E$ is ample. For $0 < \epsilon < 1$,

$$D + \epsilon A = (1 + 2\epsilon)H + (1 - \epsilon)E.$$

To compute $o_E(D + \epsilon A)$, write an effective representative as $G = aE + G_0$, with $E \not\subseteq \operatorname{Supp} G_0$. Intersecting with E gives

$$-(1 - \epsilon) = G \cdot E = -a + G_0 \cdot E \geq -a,$$

so $a \geq 1 - \epsilon$. Conversely, a general line avoiding p has strict transform L with $L \sim H$ and $E \not\subseteq \operatorname{Supp} L$. The divisor

$$(1 + 2\epsilon)L + (1 - \epsilon)E$$

is an effective representative with coefficient $1 - \epsilon$ along E. Therefore

$$o_E(D + \epsilon A) = 1 - \epsilon, \qquad \sigma_E(D) = \lim_{\epsilon \downarrow 0}(1 - \epsilon) = 1.$$

The line can also be chosen so that its strict transform contains no prescribed prime curve $C \neq E$. The same representative then gives

$$o_C(D + \epsilon A) = 0 \qquad (C \neq E).$$

Consequently

$$N_\sigma(D) = E, \qquad P_\sigma(D) = H, \qquad D = P_\sigma(D) + N_\sigma(D).$$

3.4 A nef abundance criterion

We now recall the following theorem from [Hu20, Lemma 2.8].

Theorem 3.4.1 (Nef abundance criterion)**.** *Let $f : X \to Z$ be a projective morphism from a normal algebraic variety to a variety, and let D be a nef/Z $\mathbb{R}$-Cartier divisor on X. Then the following conditions are equivalent:*

(1) *D is abundant over Z.*

(2) *There exist a birational model $\pi : X' \to X$, a surjective morphism $g : X' \to Y$ of smooth quasi-projective varieties over Z, and a nef and big/Z $\mathbb{R}$-Cartier divisor B on Y such that*
$$\pi^* D \sim_{\mathbb{R}} g^* B.$$

(3) *For a sufficiently general fibre F of f,*
$$o_P(D|_F) = 0$$
for every prime divisor P over F.

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